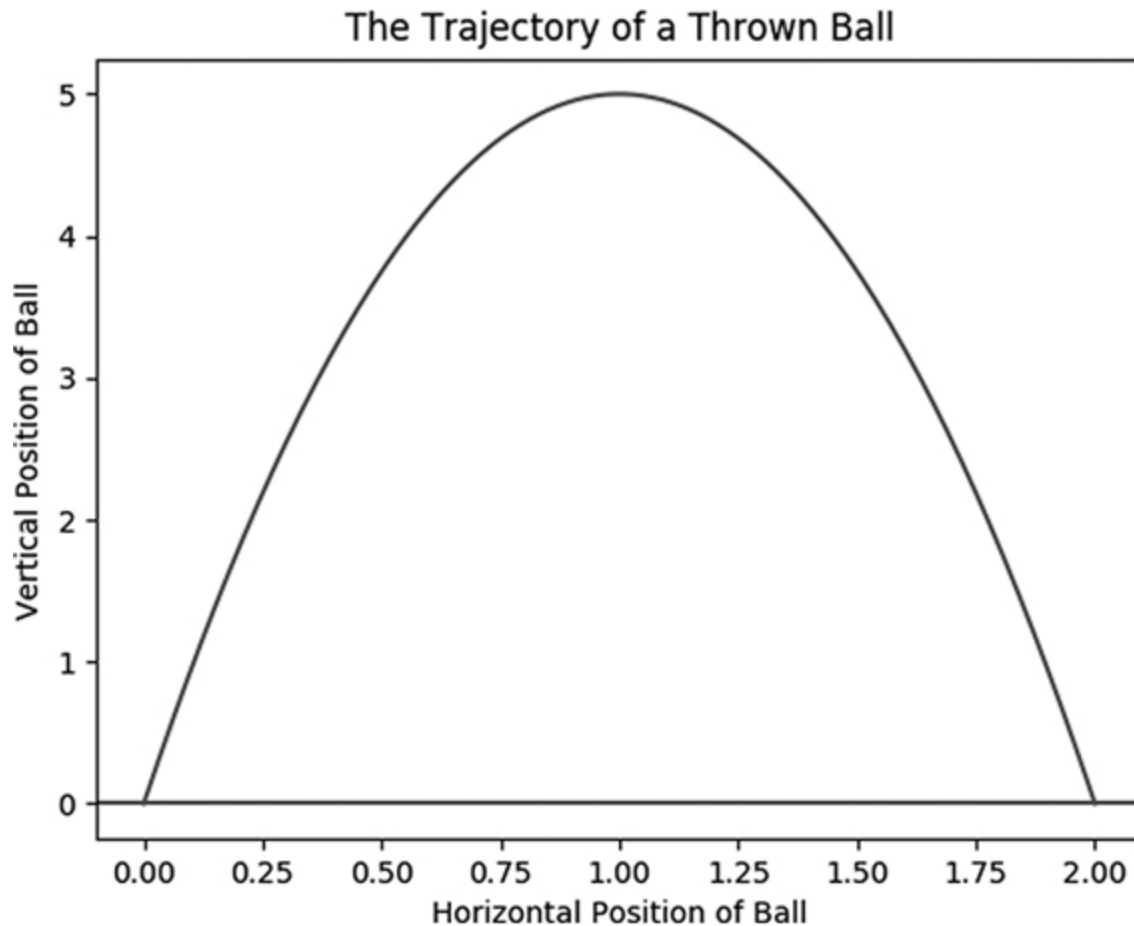




[Figure 1-1:](#) The trajectory of a hypothetical thrown ball

The Solve-for-x Strategy

Now that we have an equation for the ball's position, we can solve that equation for anything that interests us: where the ball will reach its highest point, for example, or where it will get to ground level again, which is the one thing that an outfielder needs to know in order to catch it. Students in physics classes all over the world are taught how to find these solutions, and if we wanted to teach a robot to play outfield, it would be very natural to teach the robot these equations as well. The method for solving for the ball's final location is as simple as taking the `ball_trajectory()` function we started with and setting it equal to 0:

$$0 = 10x - 5x^2$$